

Question:

What is the de Broglie wavelength of an electron with a kineticenergy of 1eV?

Answer Choices:

- A. 2.48×10^{-8} cm
- B. 5.92×10^7 cm/sec
- C. 8.66×10^{-6} cm
- D. 3.14×10^{-5} cm
- E. 9.81×10^{-4} cm
- F. 5.27×10^{-9} m
- G. 1.23×10^{-7} cm
- H. 4.14×10^{-10} m
- I. 6.625×10^{-27} erg sec
- J. 0.512×10^6 eV

Reasoning + Answer:

<think>

reasoning process here

de Broglie wavelength is given by $\lambda = h/p$, where p is the momentum and h is the Planck's constant .

Kinetic energy (KE) is given by $KE = (1 / 2) mv^2$, where m is the mass of the electron and v is the velocity .

The velocity can be calculated using KE and the mass of the electron , and then the momentum can be calculated using the velocity .

The wavelength can be calculated using the momentum and Planck's constant .

After calculations , the de Broglie wavelength comes out to be 5.27×10^{-9} m , which is equivalent to 5.27×10^{-9} m $\times 10^5$ cm/m = 5.27×10^{-4} cm .

</think>

<answer>

G

</answer>

